

Big Data Plan

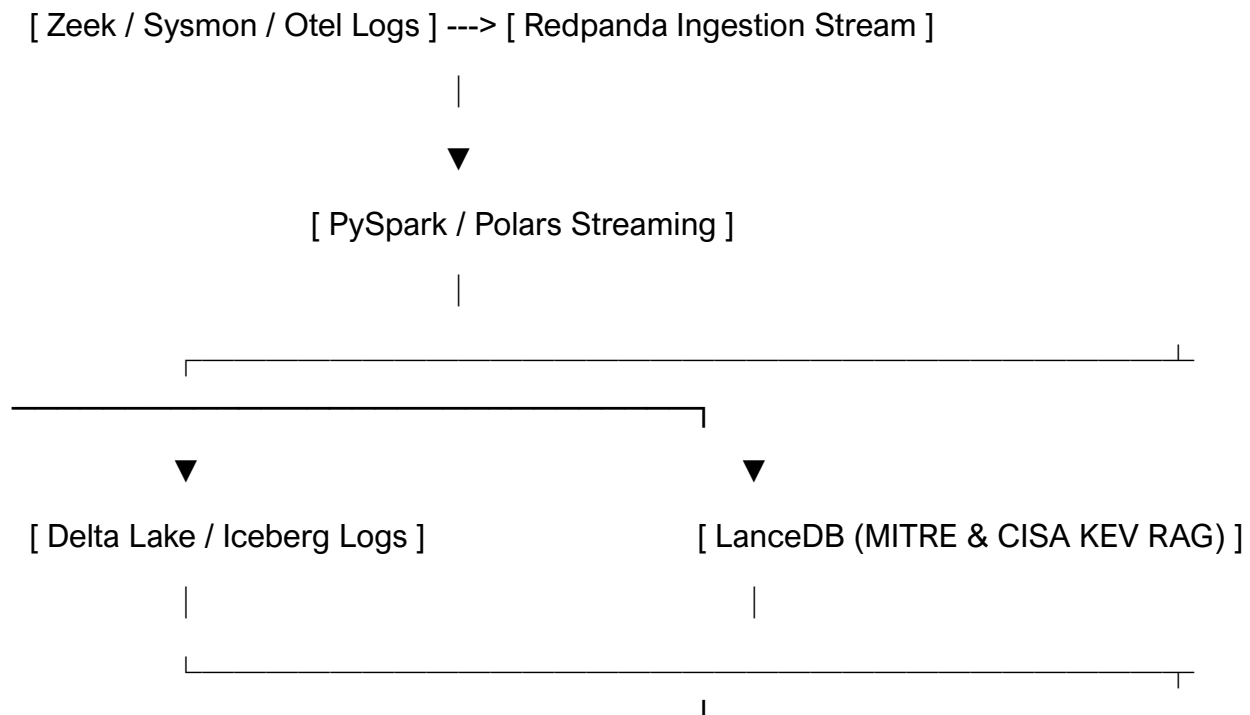
This project targets security telemetry—handling high-velocity network logs, endpoint events, and system performance metrics. You will build an autonomous, self-healing Security Operations Center (SOC) framework where AI agents continuously triage alerts, detect hidden intrusions, query threat intelligence databases, and write automated remediation rules.

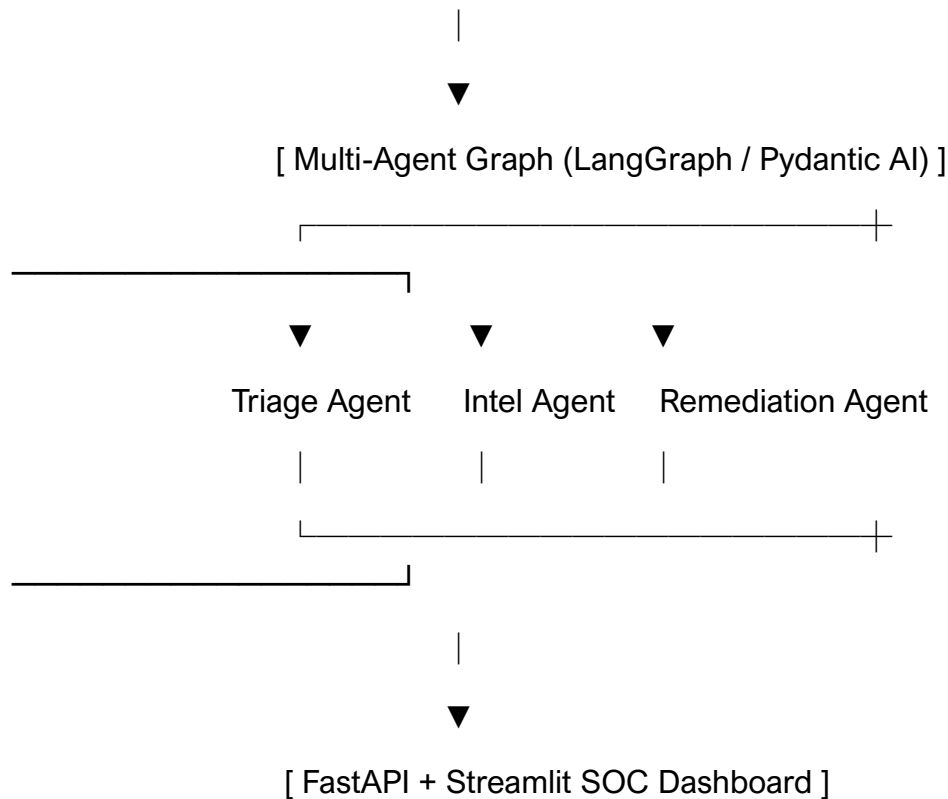
1. Project Concept & Telemetry Data Architecture

The engine continuously streams and stores system-level telemetry:

- **Network & Endpoint Logs:** Zeek connection logs (HTTP, DNS, SSL), Sysmon/Auditd host process logs, and OpenTelemetry system metrics (CPU, RAM, bandwidth spikes).
- **Threat Intelligence Data:** MITRE ATT&CK framework matrix, CISA Known Exploited Vulnerabilities (KEV) database, and emerging threat indicators.

When an anomaly occurs (e.g., DNS tunneling, unexpected SSH brute force, lateral movement, or CPU execution spikes), the system triggers an autonomous multi-agent triage graph.





2. Tech Stack Blueprint

- **Streaming & Data Lake:** Redpanda (Kafka protocol compatible), Apache Iceberg / Delta Lake, PySpark, Polars, DuckDB (for sub-second forensic log querying).
- **Machine Learning:** Scikit-Learn (Isolation Forests, DBSCAN for beaconing detection), Prophet / Statsmodels (for metric forecasting).
- **Agentic AI:** LangGraph or Pydantic AI (stateful agent execution), LanceDB / Qdrant (vector storage for threat intel RAG), Model Context Protocol (MCP) for security tool execution.
- **Observability & Guardrails:** FastAPI, Streamlit, LangSmith (agent trace auditing), Guardrails AI (to prevent unintended containment actions).

3. The 4-Week Execution Plan

Week 1: High-Velocity Telemetry Pipeline & Dual Storage

- **Goal:** Ingest streaming network and endpoint logs into structured long-term storage and vector threat indexes.
- **Tasks:**
 1. Spin up Docker containers for Redpanda, DuckDB, and LanceDB.

2. Synthesize or stream sample datasets using open telemetry logs (e.g., Zeek network logs, Sysmon host events, or Kaggle system security datasets) into Redpanda topics.
 3. Write a PySpark or Polars job to parse JSON/Syslog data into partitioned Parquet/Delta Lake tables by timestamp and event_type.
 4. Scrape the MITRE ATT&CK technique catalog and CISA KEV list, chunk the techniques/CVE descriptions, convert them into vector embeddings, and populate LanceDB.
- **Deliverable:** A streaming engine writing 500k+ structured event records per day alongside an indexed Threat Intel vector store.

Week 2: Behavioral Baselineing & Anomaly Detection Core

- **Goal:** Build statistical and ML baseline models to catch novel attacks that signature rules miss.
- **Tasks:**
 1. **Beaconing & Exfiltration Detection:** Train an unsupervised Isolation Forest / DBSCAN model on network traffic features (packet size, interval delta, egress bytes) to flag command-and-control (C2) patterns.
 2. **Volumetric Anomaly Detection:** Build a Prophet time-series model on metric aggregates (failed login attempts per minute, CPU usage spikes) to establish dynamic detection thresholds.
 3. **Data Quality Assertions:** Write continuous DuckDB SQL assertions to flag schema drift, missing protocol headers, or stream processing latency gaps.
- **Deliverable:** An ML alert engine emitting high-confidence breach indicators into an internal Redpanda alert stream.

Week 3: Agentic SOC Orchestration Engine

- **Goal:** Build a multi-agent investigation network capable of analyzing alerts, correlating evidence, and recommending actions.
- **Tasks:**
 1. **Triage Agent:** Equips a query_duckdb_logs() tool to inspect raw event logs, search for IP/user historical patterns, and evaluate alert confidence.

2. **Threat Intelligence Agent:** Equips a `vector_search_mitre()` tool to cross-reference alert behavior against MITRE ATT&CK tactics, techniques, and known CVEs.
 3. **Remediation & Guardrail Agent:** Generates actionable containment scripts (e.g., dynamic Firewall block rules, AWS Security Group updates, or Sigma detection rules).
 4. **Human-in-the-Loop Intercept:** Integrate an explicit state interrupt requiring human approval before executing any host isolation or IP blocking script.
- **Deliverable:** An autonomous multi-agent pipeline that performs forensic triage, attributes attack signatures, and drafts mitigation plans within seconds.

Week 4: SOC Control Plane UI & Agent Observability

- **Goal:** Build a SOC dashboard with real-time incident tracking, telemetry monitoring, and transparent agent reasoning logs.
- **Tasks:**
 1. Build a FastAPI backend exposing live metrics, active telemetry queries, and agent state triggers.
 2. Build a Streamlit dashboard featuring:
 - Real-time network telemetry graphs (ingestion rate, anomaly score heatmaps).
 - An interactive **Agent Execution Trace** panel displaying agent reasoning steps, tool calls, and proposed response playbooks.
 - An alert approval UI allowing human analysts to approve or reject agent containment suggestions.
 3. Integrate LangSmith to log prompt tokens, tool latency, and agent decision pathways.
- **Deliverable:** A single docker-compose up production demo environment showcasing an end-to-end autonomous telemetry defense system.

4. Key Skills You Will Master

Component	Technical Mastery Gained
Big Data Engineering	Streaming log ingestion via Redpanda/Kafka, high-throughput log partitioning in Apache Iceberg/Delta Lake, sub-second analytical querying with DuckDB.
Applied Security ML	Unsupervised anomaly detection for C2 beaconing, behavioral baselining, time-series forecasting on infrastructure metrics.
Agentic AI & RAG	Multi-agent state machine graphs (LangGraph), RAG with hybrid vector databases (LanceDB), MCP security tool execution, prompt injection guardrails.
SOC Operations	Threat attribution using MITRE ATT&CK/CISA KEV, automated Sigma rule generation, human-in-the-loop security orchestration.